

Did Consciousness Cause the Cambrian Evolutionary Explosion?

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When and where did consciousness emerge in the course of evolution? Did it happen as recently as the past million years—for example, concomitant with language or tool making in humans or primates? Or did consciousness arrive somewhat earlier with the advent of mammalian neocortex 200 million years ago (Eccles 1992)? At the other extreme, is primitive consciousness a property of even simple unicellular organisms of several billion years ago, for example, as suggested by Margulis and Sagan (1995)? Or did consciousness appear at some intermediate points? If so, where and why? Whenever consciousness first occurred, did it alter the course of evolution?

Consciousness and Evolution

According to fossil records, life on Earth originated about 4 billion years ago (Figure 37.1). For about its first 3.5 billion years (pre-Cambrian period), life seems to have evolved slowly, producing only single cells and a few simple multicellular organisms. The most significant life-forms for the first 2 billion years of this period were algae and bacteria-like prokaryotes. Then, about 1.5 billion years ago, eukaryotic cells appeared—apparently as symbiotic mergers of previously independent organelles (mitochondria, plastids) with prokaryotic cells. According to biologist Lynn Margulis (Margulis 1975; Margulis and Sagan 1995), microtubules (MT) and the dynamically functional cytoskeleton were also outside additions and originated as independent motile spirochetes which invaded prokaryotes and formed a mutually favorable symbiosis. Prokaryotic cells provided a stable, nourishing environment and biochemical energy to the spirochetes, who reciprocated by cytoskeletal-based locomotion, sensation, mitosis, and differentiation.

Pre-Cambrian eukaryotic cells created by symbiosis continued to slowly evolve for another billion or more years, resulting only in simple multicellular organisms. Then, in a rather brief 10 million years, beginning about 540 million years ago (the beginning of the Cambrian period), a worldwide, dramatic acceleration in the rate of evolution apparently occurred: the Cambrian explosion. A vast array of diversified life abruptly emerged—all the phyla from which today's animals are descended (Gould 1989).

The Cambrian explosion theory has been questioned. For example, using fossil nucleotide substitution analysis, Wray, Levinton, and Shapiro (1996) suggested a more linear process, with animals appearing about 1 billion years ago. But the more gradual, linear case assumes a constant rate of nucleotide substitution. It seems more likely that nucleotide substitution also increases during increased rates of evolution; thus, the abrupt Cambrian explosion theory still holds (Vermeij 1996).

What could have precipitated the Cambrian explosion? Were climate, atmosphere, environment, or external factors important or did a threshold of biological genetic complexity occur (Kauffman 1995; Dawkins 1989)? Can a particular biological functionality that critically enhanced adaptation, survivability, and mutation be identified? Did purposeful, intelligent behavior accelerate evolution?

The idea that behavior can directly alter genetic codes formed the basis of an eighteenth-century evolutionary theory by Jean-Baptiste Lamarck. No supportive evidence was found to show that behavior directly modified genetics, and "Lamarckism" was discredited. The question of whether behavior can alter the course of evolution *indirectly* was discussed by Schrödinger (1958), who offered several examples (Scott 1996). For instance, species facing predators and a harsh environment might best survive by producing a large number of offspring for cooperative support. Such a rapidly reproducing species is ripe for accelerated evolutionary development (Margulis and Sagan 1995). Another example is a species that finds a new habitat (moving onto land, climbing trees, etc.) to which adaptation is facilitated by supporting mutations. Changes in behavior can also favor chance mutations, which reinforce original changes and result in closed causal loops or positive feedback in evolutionary development (Scott 1996). Generally, intelligent behavior can enhance a species's survivability and enhance the opportunity for mutation to avoid extinction.

How did intelligent behavior come to be? Dennett (1995) described the "birth of agency"—the ability to perform purposeful actions—in complex macromolecules very early in the course of evolution. Dennett emphasized that agency and behavior at the macromolecular level are nonconscious and clearly preceded Cambrian multicellular organisms. For example, purposeful behavior surely occurred in unicellular eukaryotic ancestors of modern organisms such as *Paramecia* and *Euglena*, which perform rather complex adaptive movements. *Paramecia* swim in a graceful, gliding fashion via coordinated actions of hundreds of MT-based cilia on their outer surface. In this way they seek and find food, avoid obstacles and predators, and identify and couple with mates to exchange genetic material. Some studies suggest *Paramecia* can learn (e.g., they escape more quickly from capillary tubes with each subsequent attempt, Gelber 1958). Having no synapses or neural networks, *Paramecia* and similar organisms rely on their cytoskeleton for sensation, locomotion, and information processing. The cytoskeleton organizes intelligent behavior in eukaryotic cells.

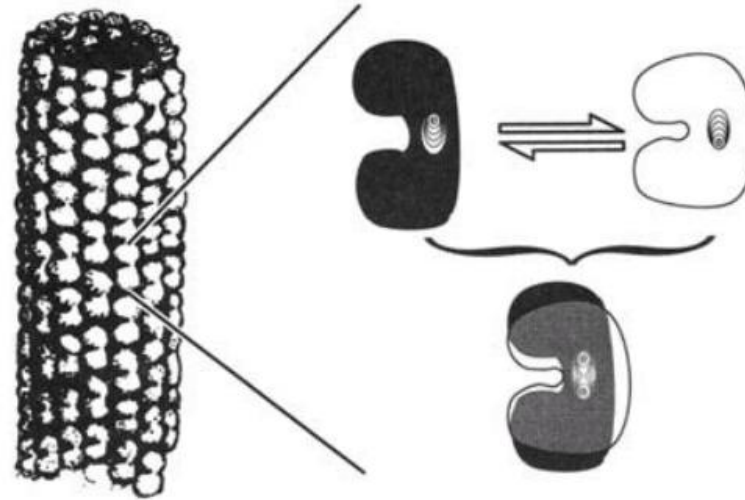


Figure 37.2

Left: Microtubule (MT) structure: a hollow tube of 25 nanometers diameter, consisting of 13 columns of tubulin dimers arranged in hexagonal lattice (Penrose 1994). Right (top): Each tubulin molecule can switch between two (or more) conformations, coupled to a quantum event such as electron location in tubulin hydrophobic pocket. Right (bottom): each tubulin can also exist in quantum superposition of both conformational states (From Hameroff and Penrose 1996).

The Cytoskeleton, Intelligent Behavior, and Differentiation

Comprising internal scaffolding and external appendages of each eukaryotic cell, the cytoskeleton includes MT, actin filaments, intermediate filaments, and complex arrays of connected MT, such as centrioles, cilia, flagella, and axonemes.

MT are hollow cylinders 25 nanometers ("nm": 10^{-9} meter) in diameter (Figure 37.2). Their lengths vary and may be quite long within some nerve processes. MT cylinder walls are hexagonal lattices of tubulin subunit proteins—polar, 8-nm, peanut-shaped dimers that consist of two slightly different 4-nm monomers (alpha and beta tubulin). MT are interlinked by a variety of MT-associated proteins to form dynamic networks that define cell shape and functions. Numerous types of studies link the cytoskeleton to cognitive processes (Dayhoff, Hameroff, Lahoz-Beltra, and Swenberg 1994; Hameroff and Penrose 1996a). Theoretical models and simulations suggest that conformational states of tubulins within MT lattices are influenced by quantum events and can interact with neighboring tubulins to represent, propagate, and process information in classic cellular automata, or ferroelectric spin-glass-type computing systems (Hameroff and Watt 1982; Rasmussen et al. 1990; Tuszynski * et al. 1995). Some authors suggest that quantum coherence may be involved in MT computation (Jibu, Hagan, Hameroff, Pribram, and Yasue 1994).

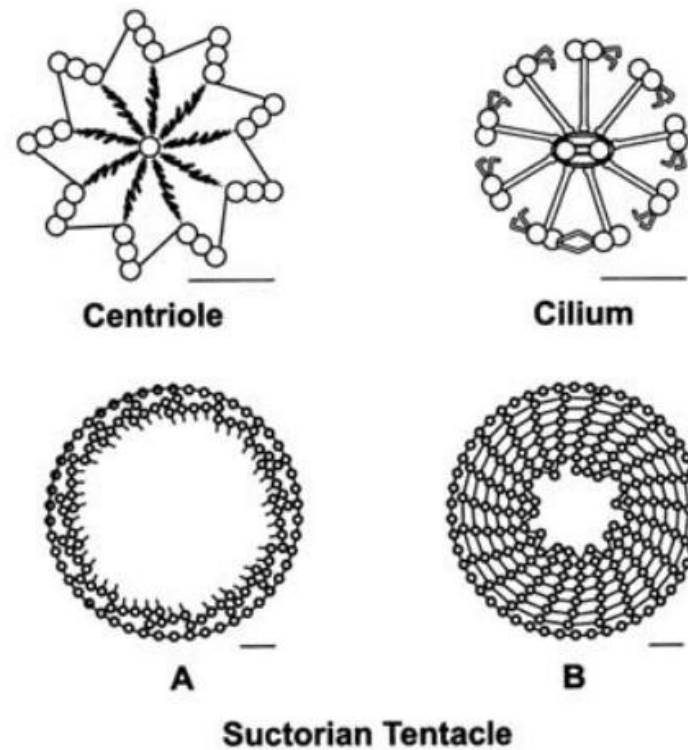


Figure 37.3

Primitive appendages comprised of microtubules (MT, shown as circles). Top left: Cross section of a centriole. Nine microtubule triplets link to a single central MT. Top right: Cilium in cross section. Nine MT doublets link to a central MT pair. (See also Insinna, Chapter 36, Figures 36.3 and 36.7). Bottom left (A): cross section of a suctorian tentacle in an open, dilated phase. Bottom right (B): suctorian tentacle in a constricted phase. Scale bars: 100nm. (Tentacle structure adapted from Hauser and Van Eys, 1976, by Dave Cantrell.)

MT are often assembled in nine MT doublets or triplets in a megacylinder found in centrioles, cilia, and flagella (Figure 37.3). Centrioles are two MT megacylinders in perpendicular array that control cell cycles and mitosis, form the focal point of the rest of the cell cytoskeleton, and provide cell navigation and orientation. Embedded in dense, electronegative material adjacent to the cell nucleus, the structural beauty, unfathomable geometry, and intricate behavior of centrioles have created an air of mystery: "biologists have long been haunted by the possibility that the primary significance of centrioles has escaped them" (Wheatley 1982; Lange and Gull 1996).

Cilia are membrane-covered megacylinders of nine MT doublets with an additional central MT pair that is both sensory and motor. Cilia can receive information and move in a coordinated fashion for locomotion, feeding, or movement of material. Flagella have the same $9 + 2$ MT arrangement as cilia but are longer and more specialized for rapid cell movement. The basic microtubule doublet $9 + 2$ (cilia, flagella, basal bodies, etc.) and $9 + 0$ (centriole) arrangements apparently originated in spirochetes prior to eukaryotes. Their cytoskeletal descendants provided agency to eukaryotes by performing a variety of purposeful behaviors.

Cytoskeletal structures also provided internal organization, variation in cell shape, separation of chromosomes, and differentiation. An essential factor in evolution, differentiation involves emergence of specialized tissues and organs from groups of cells that started out alike but began to differ and develop specific and complementary form and functions (Rasmussen 1996, personal communication). Each eukaryotic cell contains all the genes of the organism but only a subset are selected; thus, for example, liver cells are distinct from lymphocytes. Puck (Puck and Krystosek 1992) showed appealing evidence to suggest that a cell's genes are activated and regulated by its cytoskeleton, and described how differentiation requires cooperative cytoskeletal function. Tissue specialization also required factors such as actin gelation phases in cytoplasm, MT-based extensions (cilia, axonemes, etc.), and communication among cells. The most basic and primitive form of intercellular communication involves direct cell-cell channels such as gap junctions or electrotonic synapses (Lo 1985; Llinas 1985). Cytoskeletal cooperativity among neighboring cells enabled differentiation and allowed different types of tissues to emerge. Through benefit of the resultant division of labor, higher order structures (e.g., axonemes, tentacles, eye cups, nervous systems, etc.) with novel functions appeared (Rasmussen 1996, personal communication), which in turn led to more intelligent behavior in small multicellular animals.

The Dawn of Consciousness

According to this scenario tissue differentiation, agency, and intelligent behavior were slowly occurring for a billion years from the symbiotic origin of eukaryotes until the Cambrian explosion (see Figure 37.1). What happened then? Was a critical level of intelligent behavior suddenly reached? Did consciousness appear? Could primitive consciousness have significantly improved fitness and survivability beyond previous benefit provided by nonconscious agency and intelligent behavior?

One possible advantage of consciousness for natural selection is the ability to make choices. As Margulis and Sagan observed (Scott 1996) (echoing similar, earlier thoughts by Erwin Schrödinger):

If we grant our ancestors even a tiny fraction of the free will, consciousness, and culture we humans experience, the increase in [life's] complexity on Earth over the last several thousand million years becomes easier to explain: life is the product not only of blind physical forces but also of selection in the sense that organisms choose.

By itself, the ability to make choices is insufficient evidence for consciousness (e.g., computers can choose intelligently). However, noncomputable, seemingly random, conscious choices with an element of unpredictability may have been particularly advantageous for survival in predator-prey dynamics (Barinaga 1996).

Another feature of consciousness favoring natural selection could be the nature of conscious experience—qualia, our "inner life"—in the sense of Chalmers' "hard problem" (Chalmers 1996a,b). Organisms that are not conscious

but that have intelligent behavior are (in the philosophical sense) "zombies." If a zombie organism is threatened but has no experience of fear or pain, it may not react decisively. A conscious organism having an experience of fear or pain would be motivated to avoid threatening situations; one having experience of taste would be more motivated to find food. The experience of pleasure could well have promoted reproductive efforts.

Who were the early, Cambrian organisms? Fossil records have identified a myriad of small worms, strange urchins, tiny shellfish, and many other creatures (Gould 1989, e.g. depicted at the bottom of Figure 37.1). Nervous systems in small Cambrian worms (by comparison with apparent modern cousins such as the nematode worm, *Caenorhabditis elegans*) may be estimated to contain roughly hundreds of neurons. Primitive eye cups and vision were also prevalent, as were tubelike alimentary systems with a mouth at one end and an anus at the other. Cambrian urchins and other creatures also featured prominent, spinelike extensions, which seem comparable to axoneme spines in modern echinoderms such as *Actinosphaerium*. The versatile axonemes (with MT arrays more complex than those of cilia and centrioles) are utilized for sensation, locomotion, and manipulation and provide perception, agency, and purposeful, intelligent behavior.

Because consciousness cannot be measured or observed in the best of circumstances, it seems impossible to know whether consciousness emerged in early Cambrian organisms (or at any other point in evolution). The simple (hundreds of neurons) neural networks, primitive vision, purposeful spinelike appendages, and other adaptive structures that characterize early Cambrian creatures depend heavily on cytoskeletal function, and they suggest the capability for agency, intelligent behavior, and the *possibility* of primitive consciousness. Perhaps coincidentally, a specific model (Orch OR, see below) predicts the occurrence of consciousness at precisely this level of cytoskeletal size and complexity.

Consciousness and Orchestrated Objective Reduction

What is conscious experience? Believing that contemporary understanding of brain function is inadequate to explain qualia, or experience, a line of panpsychist-panexperiential philosophers (e.g., Leibniz, Whitehead, Wheeler, Chalmers) concluded that consciousness derives from an experiential medium that exists as a fundamental feature of reality. If so, conscious experience may be in the realm of space-time physics, and raw "protoconscious" information may be encoded in space-time geometry at the fundamental Planck scale (e.g., quantum spin networks, Penrose 1971; Rovelli and Smolin 1995a,b). A self-organizing, Planck-scale quantum process could select "fundamental" experience resulting in consciousness. Is there such a process?

A self-organizing quantum process operating at the interface between quantum and macroscopic states, objective reduction, is Penrose's (1989, 1994, 1996) quantum-gravity solution to the problem of wave function collapse in quantum mechanics. According to quantum theory (and repeatedly

verified experimentally), small-scale quantum systems described by a wave function may be "superposed" in different states or places simultaneously. Large-scale macroscopic systems, however, always appear in definite, classic states or places. The problem is that there is no apparent reason for the collapse of the wave function, no obvious border between microscopic quantum and macroscopic classic conditions. The conventional explanation—the Copenhagen interpretation—is that measurement or observation by a conscious observer collapses the wave function. To illustrate the apparent absurdity of this notion, Schrödinger (1935) described a now-famous thought experiment in which a cat was placed in a box into which poison is released when triggered by a particular quantum event. Schrödinger pointed out that, according to the Copenhagen interpretation, the cat would be both dead *and* alive until the box was opened and the cat observed by a conscious human.

To explain this conundrum, many physicists now believe that, intermediate between tiny, quantum-scale systems and large, cat-size systems, an *objective* factor disturbs the superposition and causes collapse or reduction to classic, definite states and locations. This putative process is called *objective reduction* (OR). One increasingly popular OR viewpoint, initiated by Károlyházy in 1966 (Károlyházy, Frenkel, and Lukacs 1986), suggests the "largeness" should be gauged in terms of gravitational effects—in Einstein's general relativity, gravity *is* space-time curvature. According to Penrose (1989, 1994, 1996), quantum superposition—actual separation (displacement) of mass from itself—causes underlying space-time to also separate at the Planck scale because of simultaneous curvatures in opposite directions. Such separations are unstable, and a critical degree of separation (related to quantum gravity) results in spontaneous self-collapse to particular states chosen noncomputably.

In Penrose's OR, the size of an isolated, superposed system (gravitational self-energy E of a separated mass) is inversely related to the coherence time T according to the uncertainty principle

$$E = \hbar/T$$

where $\hbar$ is Planck's constant over 2 and T is the duration of time for which the mass must be superposed to reach the quantum gravity threshold for self-collapse.

Large systems (e.g., Schrödinger's cat, roughly 1 kg) would self-collapse very quickly, in only 10^{-37} seconds. An isolated, superposed, single atom would not self-collapse for 10^6 years. Somewhere between those extremes are brain events in the range of tens to hundreds of milliseconds. A 25-msec brain event (i.e., one occurring in coherent 40-Hz oscillations) would require nanograms (10^{-9} gm) amounts of superposed neural mass.

In the Penrose-Hameroff Orchestrated Objective Reduction (Orch OR) model (Penrose and Hameroff 1995; Hameroff and Penrose 1996a,b), quantum coherent superposition develops in MT subunit proteins (tubulins)

within brain neurons and glia. The quantum state is isolated from environmental decoherence by cycles of actin gelation and connected among neural and glial cells by quantum tunneling across gap junctions (Hameroff 1996). When the quantum-gravity threshold is reached according to $E = \hbar/T$, OR abruptly occurs. The prereduction, coherent superposition (quantum computing) phase is equated with preconscious processes, and each instantaneous OR corresponds with a discrete conscious event.

Sequences of events give rise to a "stream" of consciousness. MT-associated proteins "tune" the quantum oscillations and the OR is thus "orchestrated" (Orch OR). Each Orch OR event selects noncomputably the MT subunit states that classically regulate synaptic-neural functions. Because the superposed protein mass separation is also a separation in underlying space-time geometry, each Orch OR event selects a particular fundamental experience.

For example, consider a low-intensity, conscious sensory perception such as activity in sensory cortex after lightly touching a finger. Such an event was shown by Libet, Wright, Feinstein, and Pearl (1979) to have a preconscious time of 500 msec until conscious awareness. For $T = 500$ msec of quantum coherent superposition, $E =$ self-collapse of approximately 10^9 tubulins. As typical neurons contain about 10^7 tubulins, Orch OR predicts involvement of roughly 10^2 to 10^3 neurons (interconnected by gap junctions) for rudimentary events. For more intense conscious events, for example, events consistent with 25-msec "cognitive quanta" defined by coherent 40-Hz activity (Crick and Koch 1990; Llinas 1985; Joliot, Ribary, and Llinas 1994), superposition and self-collapse of 2×10^{10} tubulins (and 10^3 – 10^4 neurons) would be required.

How might Orch OR have happened? One possibility is that quantum coherence emerged in eukaryotic MT assemblies via the Fröhlich mechanism as a by-product of coordinated dynamics and biochemical energy (Fröhlich, 1968, 1970, 1975).

Quantum coherence could also be an intrinsic property of the structure and geometry of MT and centrioles—introduced to eukaryotes by spirochetes. Development of actin gels provided isolation for MT quantum states, and intercellular gap junction connections (suitable for quantum tunneling) enabled larger and larger quantum states among MT in many connected cells. At some point in the course of evolution, sufficient quantum coherence to elicit Orch OR by $E = \hbar/T$ was reached. Rudimentary "conscious" events then occurred. Organisms began to have experience and to make conscious, noncomputable choices.

Three Candidates for Cambrian Consciousness

Three biological scenarios consistent with the Orch OR model for early Cambrian emergence of consciousness can be considered. Each case involves MT containing a minimum of 10^9 tubulins, sufficient for a 500-msec conscious event.

1. Sufficiently complex, gap-junction-connected neural networks (hundreds of neurons, e.g., small worms).

Many early Cambrian fossils are small worms with simple nervous systems. Hameroff and Penrose (1996b) speculated that, among current organisms, the threshold for rudimentary Orch OR conscious events (500-msec preconscious time) may be very roughly at the level of 300 neuron (3×10^9 neural tubulin) nematode worms, such as the well-studied *C. elegans*. This should be roughly the same neural network complexity as early Cambrian worms which could apparently burrow, swim, or walk the ocean floor with tentacles and spines (see Figure 1 in Gould 1989).

2. Primitive vision (ciliated ectoderm eye cup, e.g., small worms).

Another candidate for the Cambrian emergence of Orch OR consciousness involves the evolution of visual photoreceptors. Amoeba respond to light by diffuse sol-gel alteration of their actin cytoskeleton (Cronly-Dillon and Gregory 1991). *Euglena* and other single cell organisms have localized "eye spots," that is, regions at the root of the microtubule-based flagellum. Cytoplasm may focus incident light toward the eye spots and pigment material shields certain angles to provide directional light detection (Insinna, Chapter 36). *Euglena* swim either toward or away from light by flagellar motion. Having no neurons or synapses, the single-cell *Euglena*'s photic response (sensory, perceptive, and motor components) depends on MT-cytoskeletal structures.

Mammalian cells, including those of humans, can respond to light. Albrecht-Buehler (1994) showed that single fibroblast cells move toward red or infrared light by utilizing their MT-based centrioles for directional detection and guidance ("cellular vision"); he also pointed out that centrioles are ideally designed photodetectors (Figure 37.4). Jibu et al. (1994, Jibu, Pribam, and Yasue 1996) predicted that cellular vision depends on a quantum state of ordered water in MT inner cores (Figure 37.5). They postulated a nonlinear quantum-optical effect, "superradiance," that conveys evanescent photons by a process of "self-induced transparency" (the optical analogue of superconductivity). Hagan (1995, personal communication) observed that cellular vision may have provided an evolutionary advantage for single-cell organisms, with cilia, centrioles, or flagella capable of quantum coherence.

In simple multicellular organisms, eyes and visual systems began with groups of differentiated, light-sensitive ciliated cells that formed primitive eye cups (with as many as 100 photoreceptor cells) in many phyla including flatworms, annelid worms, molluscs, crustacea, echinoderms, and chordates—our original evolutionary branch, according to Cronly-Dillon and Gregory (1991).

The retinas in human eyes include over 10^8 rod and cone photoreceptors, each comprised of an inner and outer segment connected by a ciliated stalk. Because each cilium comprises about 300,000 tubulins, human retinas contain about 3×10^{13} tubulins per eye. Retinal rods, cones, and glia are interconnected by gap junctions (Leibovic 1990). Conventional vision science

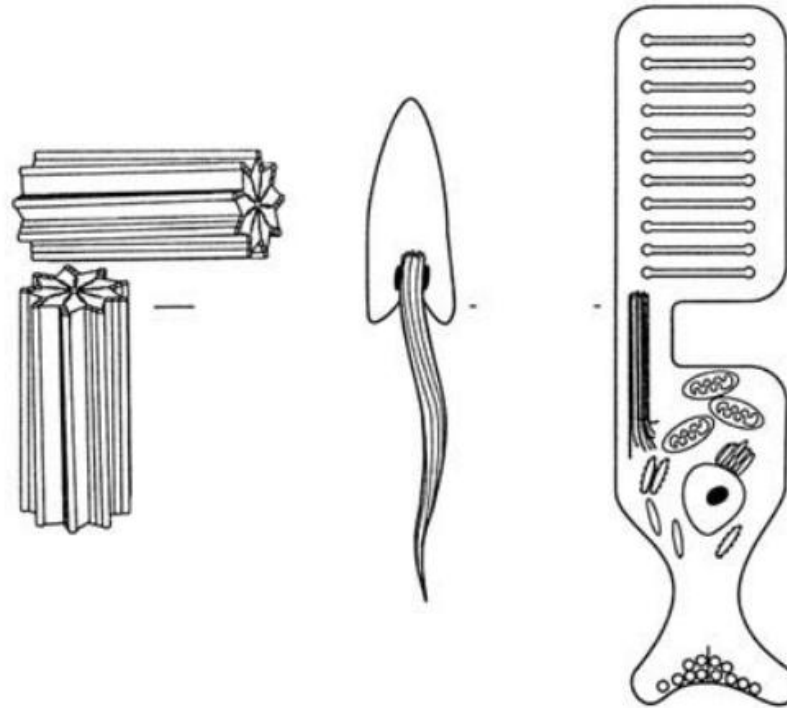


Figure 37.4

Photoreception/phototransduction mechanisms at all stages of evolution involve the nine MT doublet or triplet structures found in centrioles, cilia, flagella, and axonemes. Left: The centriole is a pair of MT-based megacylinders arrayed in perpendicular (Lange and Gull 1996). Albrecht-Buehler (1994) identified centrioles as the photoreceptor/phototransducer in photosensitive eukaryotic cells. Middle: Flagellar axonemes are the photosensitive structures in protozoa such as *Euglena gracilis* (see Insinna, Chapter 36). Right: Cilia in rod and cone retinal cells in vertebrate eyes (including humans) bridge two parts of the cells. Photosensitive pigment (rhodopsin) is contained in the outer segment (top); cell nucleus, mitochondria and synaptic connection are contained in the cell body (bottom). Light enters the eye (from the bottom in this illustration) and traverses the cell body and cilium to reach the rhodopsin-containing outer segment. Scale bars: 100 nm. (Adapted from Lange and Gull 1996 and Insinna, Chapter 36, by Dave Cantrell.)

assumes the cilium is purely structural, but the centriole-cilium-flagella MT structure, which Albrecht-Buehler analyzed as an ideal directional photoreceptor, may detect or guide photons in eye spots of single cells, primitive eye cups in early multicellular organisms, and rods and cones in human retinas. Quantum coherence leading to consciousness could have emerged in sheets of gap-junction-connected ciliated cells in eye cups of early Cambrian worms.

3. Geometrical microtubule arrays (e.g., axoneme spines in small urchins such as *Actinosphaerium*, tentacles in suctorians).

Perhaps consciousness occurred in even simpler organisms? Many Cambrian fossils are similar or related to current species that have particularly interesting MT geometrical arrangements. For example, *Actinosphaerium*

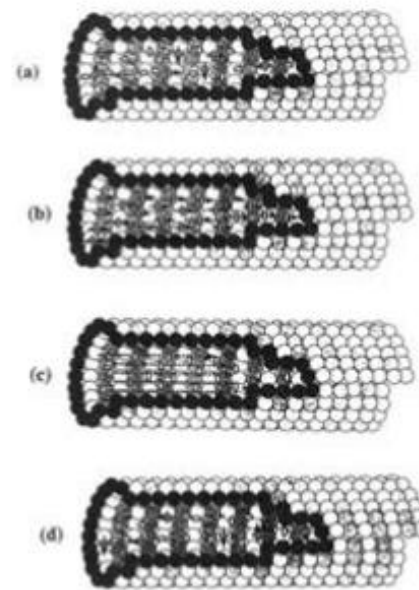


Figure 37.5

A schematic representation of the process of superradiance in a microtubule proposed by Mari Jibu et al. (1994). Each oval without an arrow stands for a water molecule in the lowest rotational energy state. Each oval with an arrow stands for a water molecule in the first excited rotational energy state. The process is cyclic ($a \rightarrow b$), and so forth. (a) Initial state of the system of water molecules in a microtubule. Energy gain caused by the thermal fluctuations of tubulins increases the number of water molecules in the first excited rotational energy state. (b) A collective mode of the system of water molecules in rotationally excited states. Long-range coherence is achieved inside a microtubule by means of spontaneous symmetry breaking. (c) A collective mode of the system of water molecules in rotationally excited states loses its energy collectively and creates coherent photons in the quantized electromagnetic field inside a microtubule. (d) Water molecules, having lost their first excited rotational energies by superradiance, start again to gain energy from the thermal fluctuation of tubulins, and the system of water molecules recovers the initial state (a). (From Jibu et al. 1994, with permission.)

(Echinosphaerium) nucleofilum is a modern echinoderm, a tiny sea-urchin heliozoan with about 100 rigid, protruding axonemes about 300 μm in length (see Figure 37.4). Appearing similar to spines of Cambrian echinoderms, *Actinosphaerium* axonemes sense and interact with environment, provide locomotion, and each comprise several hundred MT interlinked in a double spiral (Figure 37.6). Each axoneme contains about 3×10^7 tubulins, and the entire heliozoan contains 3×10^9 tubulins (Roth, Pihlaja, and Shigenaka 1970)—perhaps coincidentally, the precise quantity predicted by Orch OR for a 500-msec conscious event.

Allison and Nunn (1968) and Allison et al. (1970) studied living *Actinosphaerium* in the presence of the anesthetic gas halothane. They observed that the axoneme MT disassembled in the presence of halothane (although at anesthetic concentrations two to four times higher than that required for anesthetic effect).

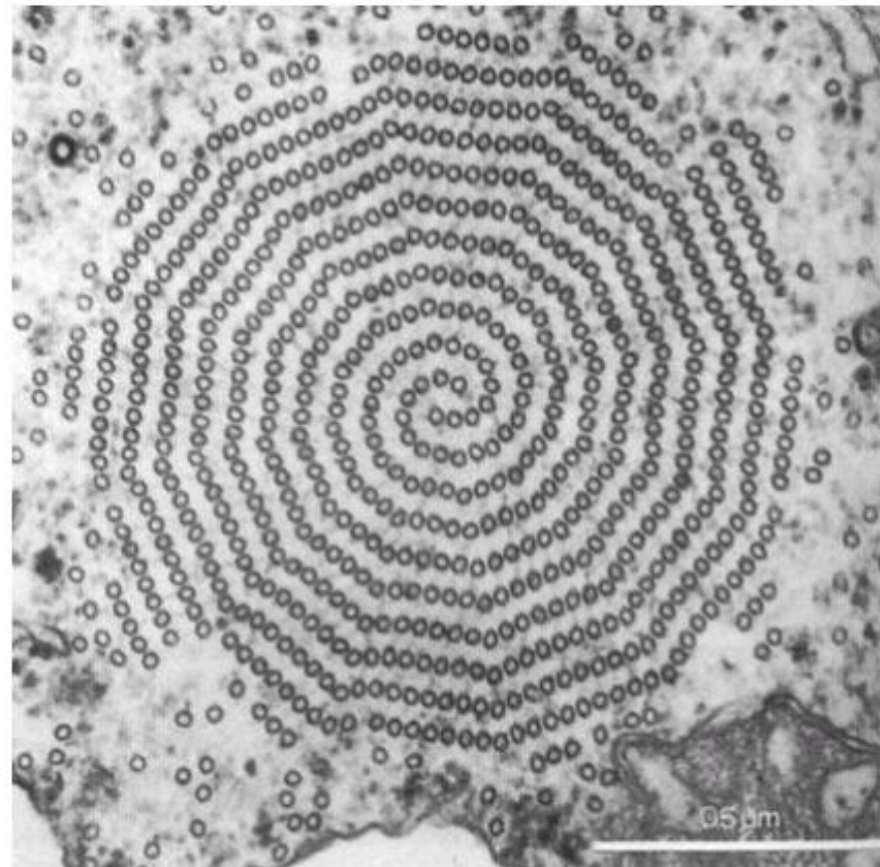


Figure 37.6

Cross-section of double spiral array of interconnected MTs in single axoneme of *Actinosphaerium*, a tiny heliozoan related to sea-urchin echinoderms present at the Cambrian evolutionary explosion (see Figure 37.1). Each cell has about 100 long and rigid axonemes, which are about 300 nm in length and are made up of a total of 3×10^9 molecules of tubulin. Scale bar: 500 nm. (From Roth 1970, with permission.)

Somewhat similar to axonemes are larger prehensile tentacles in suctorians such as *Akinetoposis* and *Heliophyra*. Small multicellular animals, suctorians (see Figure 37.3) have dozens of tiny, hollow tentacles that probe their environment and capture and ingest prey, such as *Paramecium*. The prehensile tentacles range from about 300 nm to 1 mm in length. Their internal structure comprises longitudinal arrays of about 150 MT in a ring around an inner gullet through which prey and food pass (see Figure 37.3). MT apparently slide over one another in a coordinated fashion to provide tentacle movement and contractile waves involved in food capture, ingestion, and other adaptive behaviors. The activity is interrupted by the anesthetic halothane (Hauser and Van Eys 1976). A single suctorian tentacle (150 MT, length of 500 nm) contains about 10^9 tubulins—the predicted requirement for a 500-msec Orch OR event. Perhaps consciousness arose in the probings of a Cambrian suctorian tentacle?

Would such primitive Orch OR experiences in a Cambrian worm, urchin, or suctorian be anything like experiences of humans? What would it be like to be a tentacle? A single, 10^9 tubulin, 500-msec Orch OR in a primitive system would have gravitational self-energy (and thus experiential intensity) perhaps equivalent to a human "touch lightly on the finger" experience. However, the human everyday coherent 40-Hz brain activity would correspond to 25 msec events involving 2×10^{10} tubulins, and so the typical human experience would be 20 times more intense. Humans also would have many more Orch OR events per second (40 vs. a maximum of 2) with extensive sensory processing and associative memory, which presumably, was lacking in Cambrian creatures. Nonetheless, by Orch OR criteria, a 10^9 tubulin, 500-msec Orch OR event in a Cambrian worm, urchin, or tentacle would have been a conscious experience—a smudge of awareness, a shuffle in fundamental space-time.

Conclusion

The place of consciousness in evolution is unknown, but the actual course of evolution may offer a clue. Fossil records indicate that animal species as we know them today—including conscious humans—arose from a burst of evolutionary activity some 540 million years ago in the Cambrian explosion. Thus, I suggest that:

1. Occurrence of consciousness most likely accelerated the course of evolution.
2. Small worms, urchins, and comparable creatures reached critical biological complexity for emergence of primitive consciousness at the early Cambrian period 540 million years ago.
3. Cooperative dynamics of MT, cilia, centrioles, and axonemes were the critical biological factors for consciousness.
4. Cytoskeletal complexity available in early Cambrian animals closely matches criteria for the Penrose-Hameroff Orch OR model of consciousness.
5. Orch OR caused the Cambrian explosion.

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